

Data Annotation in ASP.NET MVC (Model Validation)

- Detailed Notes

Below is a comprehensive table of data annotations available in ASP.NET MVC for model validation:

Basic Validation Attributes

Attribute	Description	Example
Required	Specifies that a property is mandatory	<code>[Required] public string Name { get; set; }</code>
StringLength	Specifies minimum and maximum length for a string	<code>[StringLength(50, MinimumLength=3)]</code>
Range	Specifies numeric range constraints	<code>[Range(1, 100)]</code> or <code>[Range(typeof(DateTime), "1/1/2020", "12/31/2020")]</code>
RegularExpression	Validates against a regex pattern	<code>[RegularExpression(@"^[A-Z]+[a-z A-Z]*\$")]</code>
Compare	Compares two properties	<code>[Compare("Password")] public string ConfirmPassword { get; set; }</code>
MaxLength	Specifies maximum length	<code>[MaxLength(100)]</code>
MinLength	Specifies minimum length	<code>[MinLength(5)]</code>
EmailAddress	Validates email format	<code>[EmailAddress]</code>
Phone	Validates phone number format	<code>[Phone]</code>
CreditCard	Validates credit card format	<code>[CreditCard]</code>
Url	Validates URL format	<code>[Url]</code>
FileExtensions	Validates file extensions	<code>[FileExtensions(Extensions = "jpg,png,gif")]</code>

Data Type Specific Attributes

Attribute	Description	Example
DataType	Specifies the data type for more specific validation	<code>[DataType(DataType.Password)]</code>
DisplayFormat	Specifies display format	<code>[DisplayFormat(DataFormatString = "{0:yyyy-MM-dd}")]</code>
ReadOnly	Marks property as read-only	<code>[ReadOnly(true)]</code>

Custom Error Messages

Attribute	Usage with Custom Message	Example
All attributes	Can include ErrorMessage parameter	<code>[Required(ErrorMessage = "Name is required")]</code>
	Can use placeholders	<code>[StringLength(10, ErrorMessage = "{0} must be between {2} and {1} chars", MinimumLength = 3)]</code>

MVC-Specific Validation Attributes

Attribute	Description	Example
<code>Remote</code>	Performs remote validation via AJAX call	<code>[Remote("IsUserAvailable", "Validation")]</code>
<code>BindNever</code>	Prevents binding from request data	<code>[BindNever] public string InternalCode { get; set; }</code>
<code>BindRequired</code>	Requires the property to be bound	<code>[BindRequired] public int Id { get; set; }</code>

Custom Validation Attributes

Attribute	Description	Example
<code>CustomValidation</code>	Specifies custom validation method	<code>[CustomValidation(typeof(MyValidator), "ValidateName")]</code>
Create custom attribute	Create class inheriting from <code>ValidationAttribute</code> and override <code>IsValid</code>	

Display Attributes (Not Validation but Related)

Attribute	Description	Example
<code>Display</code>	Specifies display name	<code>[Display(Name = "Full Name")]</code>
<code>DisplayName</code>	Specifies display name (alternative)	<code>[DisplayName("Full Name")]</code>
<code>DisplayOrder</code>	Specifies display order	<code>[Display(Order = 1)]</code>
<code>ScaffoldColumn</code>	Hides property from scaffolding	<code>[ScaffoldColumn(false)]</code>
<code>HiddenInput</code>	Renders as hidden input	<code>[HiddenInput(DisplayValue=false)]</code>

Validation Groups

Attribute	Description	Example
<code>Editable</code>	Specifies whether property is editable	<code>[Editable(false)]</code>

Complex Type Validation

Attribute	Description	Example
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<code>ValidateNever</code>	Excludes property from validation	<code>[ValidateNever] public string InternalNote { get; set; }</code>
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Sample Model with Multiple Annotations

```
public class UserRegistrationModel
{
    [Required(ErrorMessage = "Username is required")]
    [StringLength(20, MinimumLength = 5, ErrorMessage = "Username must be 5-20 characters")]
    [Remote("CheckUsernameAvailability", "Account")]
    public string Username { get; set; }

    [Required]
    [EmailAddress]
    public string Email { get; set; }

    [Required]
    [DataType(DataType.Password)]
    [RegularExpression(@"^(?=.*[a-z])(?=.*[A-Z])(?=.*\d){8,}$",
        ErrorMessage = "Password must be 8+ chars with uppercase, lowercase and number")]
    public string Password { get; set; }

    [Required]
    [DataType(DataType.Password)]
    [Compare("Password", ErrorMessage = "Passwords don't match")]
    public string ConfirmPassword { get; set; }

    [Range(18, 120, ErrorMessage = "Must be 18+ to register")]
    public int Age { get; set; }

    [DataType(DataType.Date)]
    [DisplayFormat(DataFormatString = "{0:yyyy-MM-dd}", ApplyFormatInEditMode = true)]
    public DateTime BirthDate { get; set; }

    [Url]
    public string Website { get; set; }

    [CreditCard]
    public string CreditCardNumber { get; set; }

    [Phone]
    public string PhoneNumber { get; set; }
}
```

These annotations help enforce validation rules at the model level, which are automatically enforced by ASP.NET MVC during model binding and can be displayed in views using `Html.ValidationMessageFor()` and similar helpers.

